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Revising green roof design methods with downscaling model of rainfall time series

V. Pons[1,2 *], T. M. Muthanna[1], E. Sivertsen[3] & J-L. Bertrand-Krajewski[2]

[1]*Department of Civil and Environmental Engineering, Norwegian University of Science and Technology (NTNU), N-7491 Trondheim, Norway*

[2] *Univ Lyon, INSA Lyon, DEEP (EA 7429), 11 rue de la Physique, F-69621 Villeurbanne cedex, France*

[3] *SINTEF AS, S.P. Andersens veg 3, N-7465 Trondheim, Norway*

**Corresponding author email: vincent.pons@ntnu.no*

Highlights

- The local event sampling provides distribution of green roof performance for extreme events.
- The model confirms the performance of green roofs with a larger range of hyetographs.
- The method suggests including evaluation of performance under failure in the guidelines.

Introduction

Green infrastructure and green roofs for stormwater management are often designed using the rational or variational method (Kristvik et al., 2019; Trondheim Kommune, 2015) based on Intensity Duration Frequency (IDF) curves. In Norway, the design philosophy is based on the three steps approach (3SA) (Fratini et al., 2012). There are discussions about the threshold or return period to apply on each step (Paus, 2018). Indeed, although some green roofs can be used to attenuate high return period events to some extent, they are usually not designed to cope with them. For stormwater management, the benefits of green infrastructure lie mainly in retention potential and evapotranspiration, especially in warm countries, and in the efficient use of urban space.

This study aims to improve the robustness of green roof design strategies, specifically, performance assessment for future climatic conditions using a downscaling model of rainfall time series, by addressing the following aspects: i) Sampling local extreme events to evaluate the distribution of performance depending on the return period, ii) Studying the green roof behaviour under failure, iii) Comparing performance evaluation practice with single (or low number) event to long continuous timeseries and Monte-Carlo simulations of extreme events.

Methodology

Downscaling model and event sampling strategy

A robust downscaling model based on multiplicative random cascades was developed to generate rainfall time series for different cities in Norway to be applied as future climate, with the hypothesis of the RCP8.5 scenario. The model was used together with IDF curves to generate random extreme events for each return period (Local Event Sampling, LES). The depth corresponding to a day of precipitation with each return period was downscaled from 1 day to a 6-minute timestep. The process was repeated to obtain 10000 hyetographs for each return period.

Green roof model

Two green roofs were modelled in this study: an extensive green roof (E-green roof) and a detention-based extensive green roof (D-green roof). The model was a non-linear reservoir. In order to increase the robustness of the model and its range of validity under extreme events, the model was calibrated using data from extreme tests previously performed on a large scale pilot roof (Hamouz et al., 2020).

Performance evaluation

The regulation for the city of Trondheim (Trondheim Kommune, 2015) was used to set thresholds for appropriate comparison. It should be noted that new guidelines are in preparation in this city. The regulations for local stormwater management set the peak runoff discharge threshold to 0.48 mm/min (2.5 L/s) when connected to a separate sewer system and an area of 500 m^2 and 0.33 mm/min (1 L/s) for a combined sewer system and an area of 300 m^2 with regards to a 20-year return period rain event.

Three performance evaluation strategies were used:

- The variational method (VM) (Alfieri et al., 2008) to account for strategies with a low number of events. It consists in using the constant intensity rainfall leading to the worst peak runoff according to each intensity duration frequency (IDF) curve.
- A continuous simulation (CS) to evaluate performances based on runoff distribution and estimate the mean annual duration of runoff above threshold and account for natural variation of the climate. It also allows for evaluating the annual retention fraction. A 29-year long time-series was used for this simulation.
- Local Event Sampling (LES) to sample a large number of probable hyetographs according to the location and downscaling model properties. It allows to estimate the probability to cope with a return period rain under future climate conditions in accordance with the guidelines.

Results and discussion

Figure 1 represents the comparison between the variational method (VM) and the local event sampling (LES) for current climate and projections according to the RCP8.5 scenario. The variational method estimates one single point of the distribution of the LES. In the case of the detention-based extensive green roof, VM tends to estimate a peak runoff lower than the mean of the LES distribution. Similar results were found for the extensive green roof in the case of 2-year return period events.

The detention based green roof has very stable performances (i.e low deviation) for low return period but the deviation increases with higher return period and climate change. 95% of the distribution are lower than the threshold values under current climate but the risk of runoff exceeding the threshold increases significantly with climate change and high return periods

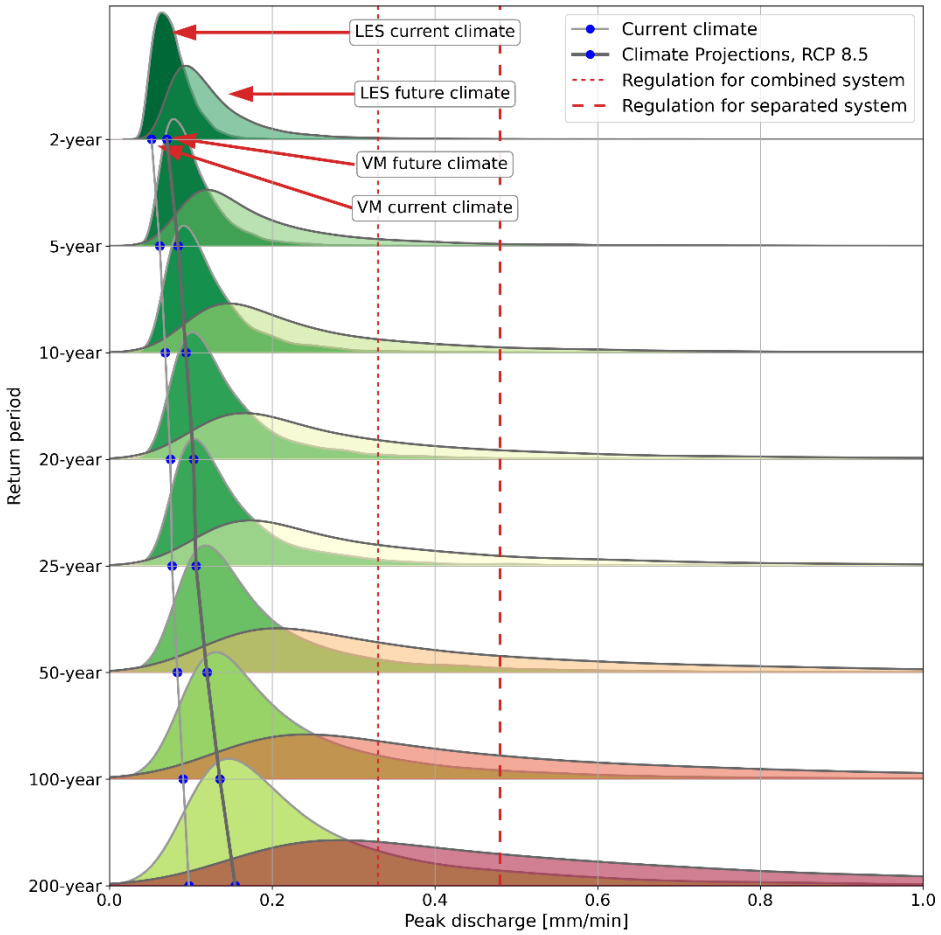


Figure 1. Predicted peak runoff of the detention-based extensive green roof using variational method (thick grey line with blue dot markers) and local event sampling (distributions) under current climate (light grey) and projected climate RCP 8.5 (dark grey).

Table 1 shows the probability to reach the target depending on the roof. For example, the probability to reach the target with RCP 8.5 and 200-year return period is only 0.33 with LES against 1 with VM. Table 1

highlights that the variational method does not allow to take an informed decision. Moreover, the estimate with the variational method does not depend on the location. The LES method provides more robust estimates because a larger variety of events likely to occur are sampled. It also includes the events able to trigger high runoff discharge in each roof. For instance, in previous studies (Hamouz et al., 2020) the D-green roof was found sensitive to specific hyetographs whereas that variational method does not include such hyetographs, which leads to less representativeness for VM. As a comparison (results not shown in the abstract), the continuous simulation (CS) allows to estimate mean annual runoff duration above threshold which would be less than 4 minutes per year in the case of the E-green roof, and 0 minutes for the D-green roof. The CS method depends on data availability but allows estimates to not depend on IDF curves or events and directly estimate frequency of exceeded threshold.

Table 1. Probability to have an extreme event depending on the green roof, the return period and the method used. The variational method (VM) can only provide a Boolean estimate.

Threshold 0.33 mm/min	E-green roof				D-green roof			
	Current period		RCP 8.5		Current period		RCP 8.5	
	LES	VM	LES	VM	LES	VM	LES	VM
2-year	0.45	1	0.14	1	1	1	0.98	1
5-year	0.26	0	0.04	1	1	1	0.88	1
10-year	0.16	0	0.01	1	0.98	1	0.79	1
20-year	0.10	0	0.005	1	0.96	1	0.68	1
25-year	0.09	0	0.003	1	0.95	1	0.65	1
50-year	0.05	0	0.001	1	0.90	1	0.53	1
100-year	0.03	0	0	1	0.84	1	0.41	1
200-year	0.01	0	0	1	0.77	1	0.33	1

Conclusions and future work

- Future climate conditions (RCP8.5) will significantly affect the runoff from green roofs, with an effect increasing with the return period.
- The LES method was found to provide reliable estimates for informed decision while the variational method remain limited in terms of design criteria.
- The use of continuous simulation might provide estimates with different nature such as mean annual runoff duration above threshold but does require long timeseries of precipitation and does not focus on extreme events.
- The results suggest to use a combination of i) LES for a risk-based approach and quantify the functioning of the roofs under failure and ii) continuous timeseries for green roof behaviour in operating state and to quantify other benefits (such as retention, attenuation of small precipitation, etc.).
- A multicriteria decision making tool needs to be applied to study the value of distribution-based estimates.

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